Aadithya S
Embedded Systems Engineer

Chennai, TN | adithyaadhi042@gmail.com | +919344469081

Summary

A passionate Embedded Systems Engineer with a strong foundation in Electronics, Bare metal and RTOS based firmware development, Web, Android and Windows Application development, seeking opportunities to apply my skills in designing innovative solutions and driving technological advancements in embedded systems.

Education

MBA –Technology Management *June 2026 – Present*
Anna University CDOE, Chennai

ME – Applied Electronics (CGPA – 8.6) *June 2024 – June 2026*
Sriram Engineering College, Chennai

BE – Electronics and Communication Engineering (CGPA – 8.9) *June 2020 – June 2024*
Sriram Engineering College, Chennai

Experience

EMBEDDED SYSTEMS ENGINEER

Phoenix Process Automation Pvt Ltd (March 2024 – Present) *Ambattur I.E, TN*

Wiring Harness Navigation System(WHNS): Designed multi-MCU hardware (**LPC1768, STM8S003, STM32F302, MSP430FR**) integrating LED matrix, Continuity testing and **VCNL4040** proximity sensing; developed firmware using **MCUXpresso, STVD, STM32CubeIDE** and **CCS** with **RS485/I²C**, also developed C# .NET software application.

Industrial Connect Innovation & Automation (Consultant) *Chennai TN*

Smart RTMS: Developed an Real-Time textile industry monitoring system using **WFI32E03UE**, 3.5-inch TS, **WBZ350-I**, **YRM1001** RFID and **DE2120** QR modules, integrating **MQTT**, **LVDS** and UART communication.

Padmaseetha Technologies Pvt Ltd (Consultant) *Chennai TN*

AED Driver & Controller: Designed **STM32F103**-based 2 kV biphasic defibrillator circuitry using IGBT/MOSFET switching, alongside DAC-based ECG simulation and differential ECG signal acquisition.

Portable Peritoneal Dialysis System: Developed an **nRF52840**-based controller integrating RTC, Hall Effect sensors, Infrared optical sensing and BLE connectivity with a Kotlin-based Android application.

Skills

- **Programming Languages**- C, C++, C#, JavaScript, VB, Kotlin, Python, Verilog HDL.
- **Schematic Design and PCB Layout Design.**

Tools

- **Schematic Design, PCB Layout Design and Simulation** - OrCAD Capture, Cadence Allegro, KiCad, Proteus
- **Embedded Development** - CCS, MCU Xpresso, STVD, Microchip Studio, MPLABX IDE, STMCube, ESP-IDF.
- **Saleae Logic Analyzer, Digital Storage Oscilloscope, Multimeter.**
- **Database** - MS SQL Server, Mongo DB, Postgre SQL, SQLite, Firebase.
- **Desktop App Development** - NET C#(Windows), Avalonia(Linux), Qt / Tkinter (Python Cross Platform)
- **Mobile App Development** - Android Studio.
- **Communication Protocols** - UART, SPI, I2C, MODBUS, CANBUS, SWIM, SWD, PROFINET, USB, 1-wire protocols.